Technical Specifications for armoured/unarmoured Underground Fibre Optic Cable, HDPE duct & RCC chambers

1. This section describes the functional requirements, major technical parameters and Type testing and Factory Acceptance Testing requirements for underground fibre optic cables. Marking, packaging and transportation requirements have also been described.

2. General

The underground fibre optic cable shall be an armoured cable suitable for underground installation in pipes. The cable should be of low weight, small volume and high flexibility. The mechanical design and construction of each unit shall be inherently robust and rigid under all condition of operation, adjustment, replacement, storage and transport.

2.1 Applicable Standards

The following standards and codes shall be generally applicable to the equipment and works supplied under this Contract:

i) ITU-T/CCITT Recommendations G.652

ii) Electronic Industries Association, EIA/TIA 455-78A, 455-3A, 455-62A, 455-164A/167A/174, 455-168A/169A/175A, 455-176, 455-59, EIA/TIA 598, EIA 455-104.

iii) International Electrotechnical Commission standards, IEC60304, IEC60794-1-2, IEC60811-5-1.

iv) Bellcore GR-20

2.2 Fibre Type(s) and Counts

The cable shall contain 24/48/96 no. of Dual Window Single Mode (DWSM) fibres conforming to G.652 as per the Technical parameters stipulated in this document.

2.3 Optical Characteristics

The attenuation coefficient for wavelength between 1525nm and 1575nm shall not exceed the attenuation coefficient at 1550nm by more than 0.05dB/km. The attenuation coefficient between 1285nm and 1330nm, shall not exceed the attenuation coefficient at 1310nm by more than 0.05dB/km. The attenuation of the fibre shall be distributed uniformly throughout its length such that there are no point discontinuities in excess of 0.1dB. The fibre attenuation characteristics specified in table 2-1 shall be "guaranteed" fibre attenuation of any & every fibre reel. Further the average cabled fibre attenuation, averaged over 100kms of cabled fibre, (as measured during the factory acceptance testing) shall be as specified in Table 2-1.

DWSM fibres shall conform to the requirements specified in Table 2-1 below:

Table 2-1

DWSM Optical Fibre Characteristics

Fibre Description:	Dual-Window Single-Mode
Mode Field Diameter:	9.2±0.4µm @1310nm 10.4±1.0µm @1550nm
Cladding Diameter:	125.0 µm ± 1 µm
Core-Clad concentricity error	≤ 1 µm
Cladding non-circularity	≤ 1%
Cable Cut-off Wavelength λcc	≤ 1260 nm
1550 nm loss performance	As per G.652 D
Proof Test Level	≥ 0.69 Gpa
Attenuation Coefficient:	@ 1310 nm ≤ 0.35 dB/km @ 1550 nm ≤ 0.21 dB/km
Chromatic Dispersion; Maximum: Zero Dispersion Wavelength: Zero Dispersion Slope:	≤ 18ps/(nm x km) @ 1550 nm ≤ 3.5 ps/(nm x km) 1288-1339 nm ≤ 5.3 ps/(nm x km) 1271-1360 nm 1300 to 1324 nm ≤ 0.092 ps/(nm²xkm) maximum
Polarization mode dispersion coefficient	≤ 0.2 ps/km^½
Temperature Dependence:	Induced attenuation ≤ 0.05 dB (-60ºC - +85ºC)
Bend Performance:	@ 1310 nm (75±2 mm dia Mandrel), 100 turns: Attenuation Rise ≤ 0.05 dB @1550 nm (30±1 mm radius Mandrel), 100 turns; Attenuation Rise ≤ 0.05 dB @ 1550 nm (32±0.5 mm dia Mandrel, 1 turn; Attenuation Rise ≤ 0.50 dB

2.4 General Construction

The optical cable shall consist of a central fibre optic unit protected by one or more layers of helically wound anti-hygroscopic tape or yarn. The central fibre optic unit shall be designed to house and protect the fibres from damage due to forces such as crushing, bending, twisting, tensile stress and moisture, wide temperature variations, hydrogen evolution etc. The fibre shall be of loose tube construction. The inner polyethylene jackets shall be free from pin holes, joints splits or any other defects. All fibre optic cable shall have a minimum service life span of 25 Years.

2.5. Colour Coding & Fibre Identification

Individual optical fibres within a fibre unit, and fibre units shall be identifiable in accordance with EIA/TIA 598 or IEC 60304 or Bellcore GR-20 colour-coding scheme. The colour coding system shall be discernible throughout the design life of the cable. Colouring utilized for colour coding optical fibres shall be integrated into the fibre coating and shall be homogenous. The colour shall not bleed from one fibre to another and shall not fade during fibre preparation for termination or splicing. Each cable shall have traceability of each fibre back to the original fibre manufacturer's fibre number and parameters of the fibre. If more than the specified number of fibres are included in any cable, the spare fibres shall be tested by the cable manufacturer and any defective fibre shall be suitably bundled, tagged, and identified at the factory by the vendor.

2.5.1. Strength Members

The central fibre optic unit should include a central strength member of Fibre Reinforced Plastic (FRP) or other suitable material. Peripheral strength members and aramid yarns are also acceptable. The central FRP strength member may be of slotted type with SZ lay (reverse oscillation lay) of fibre units or it may be cylindrical type with helical lay of fibre units.

2.5.2. Filling Compound

The interstices of the central fibre optic unit and cable shall be filled with a suitable compound to prohibit any moisture ingress or any longitudinal water migration within the fibre optic unit or along the fibre optic cable. The water tightness of the cable shall meet or exceed the test performance criteria as per **IEC60794-1-2-F5**. The filling compound used shall be a non-toxic homogenous waterproofing compound that is free of dirt and foreign matter, anti-hygroscopic, electrically nonconductive and non-nutritive to fungus. The compound shall also be fully compatible with all cable components it may come in contact with and shall inhibit the generation of hydrogen within the cable. The filling compound shall remain stable for ambient temperature up to +70°C and shall not drip, flow or leak with age or at change of temperature. Reference method to measure drip point shall be as per **IEC 60811-5-1** and drip point shall not be less than 70°C.

2.5.3. The Sheath / Inner jacket

The sheath shall be black, smooth, concentric, and shall be free from holes, splits, blisters and other surface flaws. The sheath shall be extruded directly over the central fibre optic unit and shall also be non-hygroscopic. The cable sheath design shall permit easy removal without damage to the optical fibres or fibre units. The sheath shall be made from good quality of weather resistant polyethylene compound (Black High Density Polyethylene- HDPE) and thickness shall be $\geq$ 1.8mm.

2.5.4 Armouring

Armouring shall be provided by corrugated ECCS Tape (Electrolytically Chrome Coated Steel) to make the cable protected from rodent and termite. The armour shall also provide the sufficient radial as well as compressive strength and the armour shall be electrically continuous and bonded to the outer sheath. The force of adhesion shall be minimum 14 Newton and shall be tested as per ASTM D- 4565 test method.

The thickness of the ECCS Tape shall not be less than 0.15mm. The tape shall be both side coated with a transparent copolymer of thickness 0.05 mm (min) so as to bond the armour to the outer jacket and make a unitary construction.

The ECCS Tape armouring corrugated transversely for lateral strength and bending flexibility shall be applied longitudinally with an overlap of 10% (MIN) over the main cable core. The corrugation over the entire length of the tape shall be uniform. The overlap portion of the cable armour shall be suitably protected for the ingress of the moisture (below the armour) by water swellable tape.

The height of corrugation shall be 0.6 mm (min) & the pitch of corrugated tape shall be 2.5 mm (max). Height and pitch of corrugation shall be measured between crest and trough base line.

The corrugated armouring of ECCS Tape shall offer excellent corrosive resistance. The specifications of ECCS tape shall be as per GR No. TEC/GR/TX/ORM-01/04/SEP-09 (Section XV)

2.5.5 The Outer Jacket / Termite protection.

A circular jacket of not less than 0.65 mm polymide-12 (Orange Nylon-12) material should be applied over the sheath as an outer jacket. The outer jacket shall have smooth finish and shall be termite resistant.

2.5.6 Rip Cord:

Suitable rip cord(s) shall be provided which shall be used to open the sheath(s) and the armour of the cable. The rip cord(s) shall be properly waxed to prevent wicking action and shall not work as a water carrier.

2.6 Mechanical Parameters & Tests

(A) **Tensile Strength:** The cable shall be of sufficient strength to withstand a load of value T(N)=9.81x2.5xW Newton or 2670N whichever is higher(where W is the mass of 1Km cable in Kg). The load shall be sustained for 10 minutes and the strain of the fibre monitored. The load shall not produce a strain exceeding 0.25% in the fibre and shall not cause any permanent damage to any constituent part of the cable. The change in optical attenuation during or after the application of the rated tensile load in accordance with **IEC60794-1-2-E1** procedure shall not exceed 0.05dB/Km both for 1310nm and 1550nm wavelength. The attenuation shall be noted before strain, during strain and after release of strain for all the fibres.

(B) **Crush test (Compressive Strength):** The cable shall withstand a compressive force of at least 4000 N, applied for at least 60 seconds between two plates of 100mm X 100mm in accordance with IEC60794-1-2-E3 procedure. This compressive load applied in accordance with **IEC60794-1-2-E3** shall not cause any permanent damage to any constituent part of the cable. The change in optical attenuation during or after the application of the compressive load shall not exceed 0.05dB both for 1310nm and 1550nm wavelength. The attenuation shall be noted before, during and after the test for all fibres.

(C) **Bend Radius:** The cable bend radius under no load shall be less than or equal to 20 times the cable diameter. The test method shall be according to the IEC60794-1-2-E11 (procedure-1). The fibres and component parts of the cable shall not suffer permanent damage when the cable is subjected to 10 cycles of wrapping and unwrapping of 4 complete turns around a mandrel of dia equal to 20 times the cable diameter. The change in optical attenuation after the test shall not exceed 0.05dB both for 1310nm and 1550nm wavelength. The attenuation shall be noted before and after the test for all fibres. Outer Jacket shall not show any cracks visible to the naked eye when examined whilst still wrapped on the mandrel.

(D) **Cable Bending test (Repeated bending):** The cable shall withstand repeated bending when tested in accordance with EIA-455-104 and shall not cause any permanent damage to any constituent part of the cable. The cable sample shall be at least 5 meters or more. The change in optical attenuation during or after the application of the repeated bending test shall not exceed 0.05dB. The attenuation shall be noted before and after the test for all the fibres. The test requirement shall be as mentioned below: -

Weight	:	5kg
Minimum distance from pulley centre to holding device	:	216mm
Minimum distance from weight to pulley centre	:	457mm
Pulley diameter	:	20 times to the cable dia
Angle of turning	:	90°
Number of cycles	:	30
Time required for 30 cycles	:	2 min.

(E) **Impact Test:** The cable shall withstand at least 10 impacts of 50N load from a 0.5 metre height with impacting surface radius of 12.5mm. The 10 impacts when applied at the same place in accordance with **IEC60794-1-2-E4** shall not cause any permanent damage to any constituent part of the cable. The change in optical attenuation during or after the application of the impact load shall not exceed 0.05dB.The attenuation shall be noted before, during and after the test for all fibres.

(F) **Torsion test:** The cable shall withstand 10 cycles of ±180□ torsion with 400N load applied on a 2m sample. This load cycle applied in accordance with **IEC60794-1-2-E7** shall not cause any permanent damage to any constituent part of the cable. The change in optical attenuation during or after the application of the torsion load shall not exceed 0.05dB for all fibres. The attenuation shall be noted before, during and after the test.

(G) **Kink test (Resistance):** When a cable of sample length 10 times the minimum bend radius as defined above is subjected to kinking, it shall not result in any fibre breakage and the kink shall disappear after normalising the cable. The change in optical attenuation after the application of the kink in accordance with **IEC60794-1-2-E10** shall not exceed 0.05dB for all the fibres.

(H) **Water ingress test (Resistance to water penetration):** The water ingress test of the cable shall meet or exceed the test performance criteria as per **IEC60794-1-2-F5** method B. Before applying the water tight seal at one end the outer jacket shall be stripped. A water-soluble fluorescent dye shall be used for testing. The duration of test shall be 24 hours. In addition after the test the cable shall be ripped open and the distance up to which water has seeped shall be noted.

(I) **Drip Test (Seepage of Filling Compound):**. For testing, a sample of 30 cm length of the cable with one end sealed by the end cap will be taken and outer jacket, sheath, binder tapes shall be removed by 5cms from open end of the sample. The filling compound will be wiped thoroughly and the sample be kept vertically with open end down ward in the oven for 24 hours at 70^0C temperature with a filter paper under the sample. The filter paper should not indicate any sign of drip or oily impression. The reference test specification shall be as per **IEC60811-5-1** to measure drip point.

(J) **Environmental Test:** Temperature cycling test shall be carried out on one drum length of the cable to ensure stability of attenuation parameter of the cable when subjected to temperature change which may occur during storage, transportation, and operation. The permissible temperature range for storage and operation will be from 20^0C to $+70^0$C. The rate of change of temperature during test shall be 1^0C per minute. The cable shall be kept for 12 hours at each of the following temperature and should follow the specification **IEC60794-1-2-F1.** Two cycles shall be performed.

TA2	: -20^0C
TA1	: -10^0C
TB1	: $+60^0$C
TB2	: $+70^0$C

The attenuation shall be measured at the end of each temperature range both at 1310nm & 1550nm.The change of attenuation of the fibre used shall be $\leq$ 0.05 dB/km for 1310 & 1550nm for entire range of temperature for all the fibres in each cycle.

(K) **Termite Resistance Test:** 3 (Three) Samples of optical fibre cables of 2(Two) meter length each shall be taken from the selected drums for Optical Fibre cable and the ends shall be sealed with metallic caps. These test samples will be sent to the reputed test lab for termite resistance test. The test Procedure and period shall be as per CAZRI, Jodhpur. All Samples shall be checked for any termite attack over the Nylone-12 jacket. The outer jacket shall be demonstrated to be termite resistant. Attack by termites shall be disregarded but termite should not penetrate or damage the Nylone-12 jacket of any sample. Observation on any damage of the cable shall be recorded.

(L) **Abrasion Test:** To be conducted as per IEC 60794-1-E2 or equivalent international test method.

(M) **Flexure Rigidity Test:** To be conducted as per ASTM D-790. The test shall not cause any permanent damage to any constituent part of the cable. The change in optical attenuation after the test shall not exceed 0.05dB/Km.The attenuation shall be noted before and after the test for all the fibres.

(N) **Figure of Eight Test:** 1000m of cable shall be uncoiled from the drum and arranged in figure of eight, each loop having a maximum dimension of 2m. It shall be possible to arrange cable in figure of 8 with relative ease and the cable shall not show any visible damages.

(O) **Cable Ageing Test:** After Environmental test the cable shall be subjected to a temperature of 85 ±2 °C for 168 hours. Cable shall then be brought to ambient temperature and stabilised for 24 hours. The change in optical attenuation after the test shall not exceed 0.05dB/Km for 1310 as well as 1550 nm wavelengths. The attenuation shall be noted before and after the test for all the fibres.

(P) **Embrittlement Test of Loose tube:** The minimum length of the test sample depends of the outside diameter of the loose tube and should be 85mm for tubes upto 2.5mm outside dia. The length of the bigger tubes should be calculated by using the following equation :

$Lo > 100 \times ((D^2 + d^2)/4)^{1/2}$
Where
Lo = Length of tube under test
D = Outside dia of loose tube.
d = inside dia of loose tube.

Both the ends of a buffer tube test sample may be mounted in a tool which is clamped in jaws of a tensile machine which exert a constant rate of movement. The movable jaw may move at a rate of 50 mm per minute toward the fixed jaw. Under load the tube will bend, so that the tube is subjected to tensile and compressive stresses. The fixture for holding the tube should be designed in a manner that the tube might bend in all directions without further loading. The tube should not get embrittled. No ink should appear on the tube upto the safe bend dia of tube (20 D) where D is the outside diameter of the loose tube. There should not be any physical damage or mark on the tube surface.

(Q) **Kink Resistance test on the loose tube:** A longer length of the loose tube is taken (with fibre and gel), a loop is made and loop is reduced to the minimum bend radius of loose tube i.e. 20 D. (where D is the outside dia of the loose tube). This test is to be repeated 4 times on the same sample length of the loose tube. No damage or kink should appear on the surface of the tube.

(R) **Drainage test for loose tube:** A tube length to 40 cm shall be cut and filled with filling gel ensuring there are no air bubbles and the tube is completely full. The filled tube is placed in a horizontal position on a clean worktop and cut 5 cm from each end so that the finished length of the sample is 30 cm. The filled tube shall be left in a horizontal position at an ambient temperature for 24 hrs. The sample tube is then suspended vertically in an environment heat oven over a weighed beaker. It is left in the oven at a temperature of 70 °C for a period of 24 hrs. At the end of the 24 hrs. period the beaker is checked and weighed to see if there is any gel in the beaker. There shall be no gel or oil in the beaker.

(S) **Check of easy removal of sheath :** The sheath shall be cut in circular way using a sheath removal tool and the about 300 mm length of the sheath should be removed in one operation. It should be observed during sheath removal process that no undue extra force is applied and no component part of the cable is damaged. It shall be possible to remove the sheath easily. Easy removal of both the outer jacket and the inner sheath shall be checked separately.

(T) **Effect of aggressive media on the cable surface (Acidic and alkaline behaviour)** The test shall be conducted as per method no. ISO175. The two test samples of the finished cable each of 600 mm in length are taken and the ends of the samples shall be sealed. These test samples are put in the PH4 and PH10 solutions separately. After 30 days these samples are taken out from the solutions and examined for any corrosion etc. on the sheath and other markings of the cables. The sample should not show any effect of these solutions on the sheath and other marking of the cable.

2.7 Cable drums, Marking, Packaging and Transport

All optical fibre cable shall be supplied on strong wooden drums provided with lagging with adequate strength, constructed to protect the cabling against all damage and displacement during transit, storage and subsequent handling during installation. **However, the exact lengths for drums to be supplied for each section of the link shall be determined by the Contractor during survey. Drum schedule shall be approved by the Employer before manufacturing the FO cable.** Both cable ends in the drum shall be sealed and shall be readily accessible. The drum shall be marked with arrows to indicate the direction of rotation. Both the ends of the cable shall be provided with pulling eye. The pulling eye and its coupling system should withstand the same tensile load as applicable to the cable. The following marking shall be done on each side of the cable drums.

i) Drum number
ii) Consignee's name and address
iii) Contractor's name and address
iv) Type of cable
v) Number of fibres
vi) Type of fibres
vii) Year of manufacturing, month & batch no
viii) Name of manufacturer
ix) Total cable length
x) Inner end marking and Outer end marking

Packing list supplied with each drum shall have all the information provided on marking on the respective cable drum and following additional information: OTDR length measurement of each fibre and Ratio of fibre and cable length.

2.7.1 Optical fibre cable marking

A suitable marking shall be applied in order to identify this cable from other cables. Marking on the cable shall be indeliable, of durable quality, shall last long and shall be applied at regular interval of one-meter length. Marking shall be imprinted and must clearly contrast with the surface and colors used must withstand the environmental influences experienced in the field. The accuracy of the sequential marking must be within $\pm$ 0.5% of the actual measured length. The sequential length marking must not rub off during normal installation. In case laser printing is used the marking shall not exceed 0.15 mm depth. The optical fibre cable shall have the following markings in every meter.

i) Type of Cable
ii) Running meter length
iii) Number of fibres
iv) Type of fibre
v) Laser symbol & caution notice
vi) Year of manufacture and batch no.
vii) Manufacturer's name
viii) APTRANSCO

2.7.2 Operating Instructions

Complete technical literature in English with detailed cable construction diagram of various sub-component with dimensions and test data of the cable shall be provided. All aspects of installation shall also be covered in the handbook.

2.8 Test and Inspection:

From each batch of cables presented by the Contractor for Factory acceptance testing, the Employer shall select random sample(s), to be tested for acceptance. The Sampling rate for the Factory acceptance tests shall be 10% of the batch size (minimum 2) for FO cable drums.

In case any of the selected samples fail, the failed sample is rejected and additional 20% samples shall be selected randomly and tested. In case any sample from the additional 20% also fails the entire batch may be rejected.

Since FAT testing provides a measure of assurance that the Quality Control objectives are being met during all phases of production, the Employer reserves the right to require the Contractor to investigate and report on the cause of FAT failures and to suspend further testing/ approvals until such a report is made and remedial actions taken, as applicable.

2.8.1 Type Testing.

The tests listed in table 2.2 shall be carried out as type tests for fibres.

TABLE 2.2 TYPE TESTS FOR OPTICAL FIBRES

S.No.	Test Name	Acceptance Criteria	Test Procedure
1	Attenuation	TS Table 2-1	EIA/TIA 455 – 78A
2	Attenuation Variation with Wavelength	TS Table 2-1	EIA/TIA 455 – 78A
3	Attenuation at Water Peak	TS Table 2-1	EIA/TIA 455 – 78A
4	Temp.Cycling		EIA/TIA 455 – 3A, 2 Cycles
5	Attenuation with Bending		EIA/TIA 455 – 62A
6	Mode Field Dia	Acceptance Criteria	EIA/TIA 455 – 164A/167A/174
7	Chromatic Dispersion		EIA/TIA 455 – 168A/169A/175A
8	Cladding Diametre		EIA/TIA 455 – 176
9	Point discontinuities of attenuation		EIA/TIA 455 – 59
10	Core – Clad connectivity error		EIA/TIA 455 – 176
End of Table			

The cable to be supplied should have been type tested as per relevant TEC specification including latest amendment. The Bidder shall submit along with their bid earlier carried out type test reports as per TEC specification (Mandatory requirement) for the offered fibre optic cable.

2.8.2 Factory Acceptance Testing

The tests listed in Table 2-2 shall be carried out as Factory Acceptance Test for Underground fibre optic cable meeting the requirements specified in this section. However, for **Water Ingress Test** and **Environmental Test, previous test reports for similar type of test conducted by any reputed/ government organisation shall be submitted instead of performing these tests.**

Table 2-3

Factory Acceptance Tests on Underground Fibre Optic Cable

S. No.	Factory Acceptance Test
1	Attenuation Coefficient (1310, 1550): By EIA/TIA 455- 78A or OTDR
2	Point discontinuities of attenuation: By EIA/TIA 455- 78A or OTDR
3	Visual Material verification and dimensional checks as per TEC specification.
4	Water Ingress test - **Previous Test Report for similar type of test conducted by any reputed/ government organisation shall be submitted.**
5	Tensile strength test / Strain test
6	Impact test
7	Kink test
8	Environmental test - **Previous Test Report for similar type of test conducted by any reputed/ government organisation shall be submitted.**
9	Crush Test
10	Drip test

3.0. Specification for Optical Fibre Joint Box

This specification gives the general requirements, type and factory testing requirements of Joint Box for optical fibre cables. The packaging and transportation requirements have also been specified.

The Joint Box should be suitable for use in Manholes as well as direct buried applications. It shall also be possible to branch out the cable from the Joint box as and when required without damaging the existing cables. The Joint box shall have minimum 4 single cable entry ports and one oval port (60x40mm) having sealed designed such that it will be opened by cutting only as and when required. All materials used for Joint box and sealing shall be termite resistant.

3.1 Construction of Joint Box

The Joint box shall be manufactured as per the latest state of art technology. The design of the Joint box must ensure:

(i) Mechanical and optical protection of the fibre.
(ii) Environmental protection of the splice.
(iii) The integrity of the connections and cable seals, fibres and buffers during mounting, functioning and dismounting.
(iv) The possibility of the repeated reopening and re-closing of the joint box and fibre organizer without removing or modifying the joint box but only by replacing the sealing.
(v) The addition of new cables shall be possible in the same joint box, if required.
(vi) It shall be possible to terminate all cables having outer diameter from 8mm to 18mm.
(vii) The size of the Joint box shall be minimum 450mm X 135mm (Height x Dia.).
(viii) The size of fibre organizer cassette shall be $\geq$ 300mm X 95mm (Length X Width).

The Joint Box shall be complete in all respect and shall consist of following main parts:

3.1.1. Main Box

The main box shall be sturdy, durable box having a base and dome shaped body. The dome shall be fixed on the base. The domed shaped body shall cover the entire junction while the base shall enable the entries of the optical fibre cable. The base and dome shall be made of thermoplastic/High density polypropylene material. The Joint Box should be suitable for opening and reentry frequently without impairing its properties. The body may have ribs as strength member if required.

3.1.2 Cable organizer (Strength member and cable termination)

Cable organizer shall be suitable to secure extra length of fibre tubes with safe bending radius. It should not cause any strain or tension on the fibre. It shall be possible to fix the strength member(s) and the optical fibre cable firmly so that the cable arrangement will not shift or move laterally inside the Joint box. The Internal structure shall be metallic (made of stainless steel) to support and hold the cables and strength members etc. The metallic parts for making connections shall be made of Brass or Nickle Chromium plated steel and the total assembly shall be corrosion proof.

3.1.3 Fibre organizer/ Fibre Splice trays

Fibre organizer shall be non-metallic made of ABS material having following characteristic.

(i) Specific gravity	1.01-1.21 gm/cc	ASTM-D-792
(ii) Tensile strength	0.002kg/sqmm - 0.005kg/mm	ASTM-D-638
(iii) Elongation	<50%	ASTM-D-638
(iv) Water absorption	0.3-0.4	ASTM-D-57-59
(v) Rock well hardness	R81-R111	ASTM-D785A

Contractor shall furnish test certificates in conformity to the above parameters of the ABS material. Fibre organiser cassettes shall be provided on which the fibre splice and service loops of fibres may be placed by making fibre coils. Slots on the splice tray for fixing splice protection sleeve shall be in such a way that they will not cause any stress or strain on sleeve or fibre and shall not shift, loose or move inside the tray or come into conflict with the fibre coils once fixed. It shall be possible to fix a minimum of 4 secondary tubes at the entry port of each tray. No PVC or any other type of adhesive tape is permitted to hold fibres and loose tube inside the tray. To the extent possible all fibres of a tube shall be spliced in a single tray for better tube identity and fibre looping. In case it is not possible to splice all fibres of a tube in a single tray, the remaining fibres of the tube shall be looped back to the adjacent tray with transport tube supplied with joint box. The quantity of splice trays in the Joint box shall be as per the number of fibres in a particular cable as mentioned below:

Type of cable splices/tray	**Splice trays (Fibre organiser)**	**Fibre**
24/48/96 Fibre	2/4	12F/24F

The fibre organizer shall be fixed inside the Joint box in such a way that this shall not loosen once fixed or to shift or move in any way.

3.1.4 Holding Arrangements

The box shall provide the following:

(i) Holding arrangement and framework for properly securing cable organizers with splice trays.
(ii) Securing arrangement for holding fibres.
(iii) Holding device to hold strength member of fibre optic cable securely.
(iv) Any other extra component required for providing strength and reliability to the Joint Box.

3.1.5 Compatibility

All the component and parts used shall be compatible with the optical fibre cable, fibre splices and cable components. Their use for long should not result in increase in transmission loss or deterioration in other properties.

3.1.6 Marking on body of the Joint box

The following information by marking on Joint box shall be provided:

(i) Manufacturer’s name & date
(ii) Type of Joint box
(iii) Number of Splice organizer cassettes
(iv) Number of splices per cassette
(v) Batch number and serial number.

3.2 General

The Joint Box shall confirm to the ITU-T Standards or latest standards prescribed by TEC.
Other important requirements are as follows.

(i) Assembly: The method of assembly, box and reopening shall be simple and adoptable with minimum training.

(ii) Component: All component shall be durable and of good quality. Number of component should be minimum and adequate.

(iii) Consumables: Consumables mentioned in Table 3.1 or any other items required shall form a part of Joint Box.

(iv) Capacity of Joint Box: The Joint Box shall be suitable for splicing for 24/48/96 fibre cables. Suitable Joint Box shall be provided to encase the optic cable splices in a protective, moisture and dust free environment. The Joint Box shall be designed for the storage and protection of a minimum of 24/48/96 optical fibre splices and equipped with sufficient number of splice trays for splicing all fibre in the cable. If and when required the capacity of Joint box can be increased by adding extra splice trays only.

3.3. Joint box Installation

Installation and sealing of Joint box shall be done according to the instruction manual provided with joint box. The following general criteria must be fulfilled.

(i) Fibre loose tubes shall be placed safely inside the joint box and routed properly.

(ii) Sheath shall be fully inside the joint box and FRP shall be properly fixed.

(iii) Silica Gel shall be placed inside the joint box before closing the box.

(iv) The sealing portion of the cable and joint box shall be filled before placing heat shrinkable sleeves. Rubber gaskets may be placed according to the instruction manual, if any.

(v) The heat shrinkable sleeves shall be sealed with very low flame of Hot gun till sealing liquid properly melt and gripped. The cable entry ports being used for cable entry into the joint box shall be sealed by heat shrink sleeves only for ease of installation and reentry. The length of the sleeve shall cover the full length of the cable entry port and at least 30mm length of the cable. Heat shrink tube shall have wall thickness of at least 1.5mm before shrinkage and $\geq$ 2.5 mm after recovery. For sealing of base with dome, contractor may use mechanical sealing or heat shrink sleeve system. In case of mechanical sealing contractor shall indicate the exact method. The gasket or the “O” ring required for sealing shall be made of Neoprene rubber and after sealing it shall be air-tight.

(vi) The Joint box shall be housed inside the manhole horizontally or vertically by suitable means.

3.4 Tests and Inspection:

3.5 Type tests:

The joint box offered for supply shall have been type tested as per TEC specification (Mandatory). The Bidder shall submit along with their bid the earlier carried out type test reports as per TEC specification.

3.6 Factory acceptance tests

Factory acceptance test shall be waived off, if the Joint box offered for supply meets the requirement as stated in section- 3.4 and section 3.5.

3.7 Packing and Transportation:

The Joint Box shall be transported after proper packing including the list of all sub-component, required maintenance spares. Consumables, mentioned in Table 3.1 shall be supplied along with each Joint box. Each Joint Box shall be fully packed with all its components. All tools and accessories should be properly packed with each Joint Box and the Joint Box packets should be placed inside an adequate size container for good looking packing and safe transportation.

Table 3.1

List of consumables required with Joint box

S.no	Item	Qty.
1	Protection sleeves For example, for 24 fibre Joint box, protection sleeves shall be 36.	As per the fibre count in the cable + 50% extra quantity
2	Cable tie	Required Number & length
3	Tissue/Lens paper 3”x4”	100 numbers
4	Acetone (Lab grade)	100 ml
5	Hexane (Lab grade)	100 ml
6	Isopropyl (Lab grade)	100 ml
7	PVC adhesive tape (5meter length of 10mm width	1 number
8	Double side adhesive tape	1 number
9	Sealing arrangement	1+1 (One complete set to be supplied as a spare) for all sealing ports.
10	Cable and fibre identification rings	2 sets
11	Transport tube	As per the requirements
12	Gasket or O-ring (in case of mechanical sealing)	1+1 (One O-ring to be supplied as a spare
13	Any other item	As per the requirements
End of Table		

4.1 Installation of the armoured/unarmoured Underground Optic Fibre cable

The cable shall be installed inside HDPE green duct already installed earlier. The cable shall be installed by compressed air blowing technique. The cable blowing machine shall be capable of blowing the fibre optic cable length of at least 1 km or more. It shall be possible to blow more than **8 km** per day using the blowing machine. The contractor shall propose the exact methods and procedures for installation taking into consideration the following guidelines for approval by the employer.

a. The Optical Fibre Cable Drums shall be handled with utmost care. The drum shall not be subjected to shocks by dropping etc. They shall not be normally rolled along the ground for long distance and when rolled, shall in the direction indicated by the arrow. The battens shall be removed only at the time of actual laying.
b. A blowing machine in association with an appropriate compressor shall be used for blowing.
c. Temporary blowing chambers (if required) shall be constructed and then backfilled after blowing operation is completed.
d. Locations along the route, which provide easy access points for blowing machine and compressor, shall be determined.
e. Before starting the cable blowing, both HDPE pipes installed under this package shall be checked for obstacles or damage. The already installed HDPE pipe wherein cable are to be installed under this package shall also be checked for obstacles or damage. Checking shall be done by using a proper sized mandrel equipped with a transmitting device.
f. Always blow downhill wherever possible.
g. Multiple blowing machines may be used in tandem if so required.

Installation by pulling may be permitted by the Employer only in specific cases where installation by blowing is not feasible. In case pulling is used, the pulling speed shall be determined considering the site condition. Care must be taken not to violate the minimum bending radius applicable for the fibre optic cable. Tension in the cable during laying shall not exceed tension limit given in Section 1 and the cable should not be damaged during or after the pulling.

While installing the cable, excess length of about 10 meters shall be stored at each joint location for each side. Excess length of 10 m shall be kept at one ends of a road crossing, culvert crossing and 20 meters at one end of bridges, However, exact excess lengths and manhole locations shall be finalised during detailed engineering. The excess length shall be housed in manholes.

Before undertaking the blowing activity, the spare duct shall be inspected for any damage and in case, the damage is noticed, it has to be rectified. The rectification work shall be in the scope of the contractor including **civil work**, without any additional cost to APTRANSCO. The **Duct integration test** shall be conducted as per APTRANSCO approved norms and practices, before taking up the blowing activity.

4.2 Installation of Joint Box.

The joint box shall be properly installed, assembled and sealed as detailed in section 3. The joint box shall be vertically installed on the wall of man hole using suitable fixtures.

4.3 Optical Fibre Splices

Splicing of the optical fibre cabling shall be minimized through careful planning. All works of splicing shall be carried out inside an air conditioned environment. It is important that all splicing work be done under very clean conditions and under controlled temperature as it contributes to the quality of the splice and prevents building of temperature-dependent fibre tension into the splice. All required splices shall be planned to occur at Joint location/manhole. All optical fibre splicing shall comply with the following:

1. All fibre splices shall be accomplished through fusion splicing.
2. Each fibre splice shall be fitted with a splice protection sheath fitted over the final splice.
3. All splices and bare fibre shall be neatly installed in covered splice trays.
4. For splicing of each fibre, every effort shall be made to minimise the bi-directional average splice loss. It is recommended that during splicing, on-line splice loss testing shall be carried out by a team with OTDR from open ends on both sides. The splicing team shall work in conjunction with the testing teams for on-line correction/rectifications of the splices.
5. Average bi-directional splice loss at any particular splice shall not exceed 0.1dB but total bi-directional average of all splices in a link shall not exceed 0.05dB.
6. Fibre optic cable service loops as indicated in Section 4.4 Service Loops shall be provided.

4.4 Service Loops.

For purposes of this specification, cable and fibre service loops are defined as slack (excess) cable and fibre provided for facilitating the installation, maintenance and repair of the optical fibre cable system.

a. Outdoor Cable Service Loops: At manhole chambers splices are installed with sufficient fibre optic cable service loops (as provided in Section 4.1) such that the recommended minimum bend radius is maintained while allowing for installation or maintenance of the cable to be performed in a controlled environment at ground level. Optical cable service loops (excess cable) shall also be provided at all crossings in manholes (as provided in Section 4.1).

b. Fibre Units Service Loops: For all fibre optic cable splicing, the cable shall be stripped back a sufficient length such that the fan-out of fibre units shall provide for at least one (1) metre of fibre unit service loop between the stripped cable and the bare fibre fan-out.

c. Fibre Service Loops: At least 0.5 metre of bare fibre service loop shall be provided on each side of all fibre splices. The bare fibre service loops shall be neatly and safely installed inside covered splice trays.

4.5 Site Acceptance Testing (SAT) for Underground Fibre Optic Cable

SAT for optical fibre cable shall be carried out link by link from permanent Manhole to permanent manhole.

Prior to installation, every spooled fibre optic cable segment shall be tested for compliance with the Pre-shipment data previously received from the manufacturer. This requirement will preclude the installation of cable segments that may have been damaged during shipment. Test requirements are as per table 4-1.

Optical fibre attenuation shall be measured after installation and before splicing. Any increase in attenuation or step discontinuity in attenuation shall not be acceptable and shall constitute a cable failure. The Contractor shall have to either replace the concerned cable span at its own cost or provide additional splicing, joint box and manholes required to rectify the fault at its own cost. The fibre attenuation shall be tested again after replacement or rectification of fault.

During the installation, spliced cable segments shall be tested and documented. In case it is found that the splices are bad (loss is unacceptable as per approved test procedures), the Contractor shall have to do re-splicing and provide new Joint Box wherever required at no additional cost to the Employer. After re-splicing the end to end testing shall be repeated. The splice testing requirements are indicated in table 4-2.

Upon completion of a continuous cable path, all fibres within the cable path shall be demonstrated for acceptance of the cable path. Test requirements are indicated in table 4-3 and in no case losses attributed due to other factors viz. extra splice, kinks, will be acceptable to the limit determine by the following formula:

Max <u>attenuation @ 1550nm:</u> 0.21dB/km + 0.05dB x total no of splices + 0.5dB x connector
Max <u>attenuation @ 1310nm:</u> 0.35dB/km + 0.05dB x total no of Splices + 0.5dB x connector As averaged over 100 km of fibre.

Table 4-1:
Fibre Optic Cable Pre-Installation Testing

Item:	Description:
1.	Physical Inspection of the cable assembly for damage
2.	Optical fibre continuity and fibre attenuation with OTDR at 1550 nm

Table 4-2:
Fibre Optic Cable Splice Testing

Item:	Description:
1.	Per splice attenuation with OTDR (bi-directional average)
2.	Physical inspection of Joint Box for proper fibre routing techniques
3.	Physical inspection of sealing techniques, weather proofing, etc.

Table 4-3:
Fibre Optic Cable Commissioning Testing

Item:	Description:
1.	Fibre continuity and link attenuation (bi-directional) between FODP connectors at two ends for each fibre at 1310 &1550 nm by OTDR
2.	Fibre continuity and link attenuation (bi-directional) between FODP connectors at two ends for each fibre at 1310 &1550 nm by Power Meter & Laser Source
3.	Average splice loss (bi-directional) for each splices and average splice loss for the link by OTDR at 1550 nm as per section 4.3.5
-End of Table-	

4.5.1 SAT for Joint Box

20% Joint Box shall be tested for water penetration. Installed joint box shall be completely immersed in water for 1 hour under 1 meter head, then opened and observed for water/moisture ingress. The Joint Box shall then be sealed and installed again. In case any water/moisture is detected, test shall be declared failed and all the Joint Boxes shall be tested, all failed joint boxes shall have to be re spliced and resealed.

4.6 Interconnection

The contractor shall interconnect their Fibre Optic Cable system at both ends with existing underground Fibre Optic and/or OPGW system through splicing in Joint boxes or through connection by patch cords at Permanent Manholes/ FODPs, as required.

GUARANTEED TECHNICAL PARAMETERS of ARMOURED UNDERGROUND FIBRE OPTIC CABLE

Sl.No.	Description	Tech. Particulars
1	Make & Model	
2	No. of Fibres in OPGW	24/48/96
3	Mode	DW-SM
4	Buffer type	Loose tube
5	Buffer tube diameter	2.5 mm/1.8 mm
6	Buffer tube material	PBTB
7	No. of buffer tubes	4/6/ 4 or 8
8	No. of fibres per tube	6/8/24 or 12
9	Identifiaction/numbering of individual tubes	blue, orange, green ,brown,slate/gray, white, Red & Black
10	Filling material	Thixotropic Jelly
11	Inner Strength members	FRP
12	Binding yarn/tape	Water swellable Yarn & Polyster tape with blinder
13	Approximate outside diameter	13.5+/-0.5.
14	Cable diameter	13.5
15	Cable cross section area	143
16	Jacket non- circularity	10
17	Rip cord provided	Yes
18	Fully compliant with IEEE P1222	Yes
19	Peripheral Strength member	Nil
	MECHANICAL	PROPERTIES OF CABLE
20	Max. Tensile Strength	4 k N
21	Fibre Strain margin	Less than 0.25
22	Weight	150+/-10 kgs / km
23	Crush strength	4100 N/10cm
24	Modulus of Elasticity	143 kg/Sq.mm
25	Minimum bending radius	180 to 200 mm (Operation)
26	Maximum bending radius	300 mm
	THERMAL PROPERTIES OF CABLE	
27	Coeffiecint of linear expansion	9.08 X 10^{-6} Per ºC
28	Coefficient of expansion Cladding Core	5.5 X 10^{-7} Per ºC
29	Nominal operating temperature range	- 20 ºC to 70 ºC
30	SC current transient peak temperature	100
31	Maximum allowable temperature for lightning strike	- 20 ºC to 70 ºC

32	Available length of per spool: Min Max	2KM+/-10% 2.2km 1.8km
33	Splice loss(Max. & Min. allowable)	0.1 dB, 0.01dB
34	Operating Temperature range	- 20 ºC to 70 ºC
35	Expected Cable Life	>30
	DUAL-WINDOW	SINGLE MODE FIBRES
36	Fibre description Mode field diameter	Dual- Window Single Mode Fibre 9.2±0.4µm @1310nm 10.4±1.0µm @1550nm
37	Core diameter	9.2 ± 0.4 µm
38	Cladding diameter Core-Clad Connectivity error	125.0 ± 1 µm ≤1.0 µm
39	Cladding noncircularity	≤ 1 %
40	Cable cut-off wave length	<1260nm
41	1550 loss performance Proof test level	As per g.652 >=0.35gpa
42	Attenuation Coefficient @1310nm - @1550nm -	 ≤0.35 d B /km ≤0.21 d B /km
43	Attenuation at water peak (1383nm)	<2.1 db/km
44	Chromatic Dispersion Co efficient Zero dispersion wavelength Zero dispersion slope	≤18ps(nm*km)@1550nm ≤3.5ps(nm*km)@1288-1339nm ≤5.3ps(nm*km)@1271-1360nm 1300-1324nm ≤0.092ps(nm2*km) Max
45	Polarization mode dispersion coefficient	≤ 0.2 ps/km^½
46	Temperature dependence	Inducted attenuation ≤0.05db(-60deg.c+85deg.c)
47	Bend performance G.655	@ 1310 nm (75±2 mm dia Mandrel), 100turns: Attenuation Rise ≤ 0.05 dB @1550 nm (30±1 mm radius Mandrel), 100 turns; Attenuation Rise ≤ 0.05 dB @ 1550 nm (32±0.5 mm dia Mandrel, 1 turn; Attenuation Rise ≤ 0.50 dB

Technical Specifications of HDPE Duct pipe ,RCC Chambers & Associated works for laying in UG-XLPE power cable trench

Name of the Work: HDPE Duct & Associated works in UG-XLPE power cable trench:

Technical Specifications:

1. The HDPE pipe shall conform to the following standards and the specifications described in the following section.

i. IS:4984 - Specifications for HDPE Pipe

ii. IS :2530 - Test method for polyethylene moulding materials and compounds

iii. IS: 14151 (part1) - Polyethylene pipes for sprinkler irrigation systems (part-1 pipes)

iv. IS: 9938 - Recommended colours for PVC insulation for LP wires and cables.

v. Tec-Spec No. - HDPE pipe for use as duct for optical fibre cable.(G/CDS-08/01/Dec-99)

vi. IS: 7328 - HDPE material for moulding and extrusion

vii. ASTM D 1693 -Test method for environmental stress cracking of polyethelene plastic

viii. IS:12235 (Para9) - Method of test for unplacticized PVC pipe for portable water supplies, impact strength at Zero Degree Centigrade

ix. ASTMD 1505 - Test method for density.

x. ASTMD 3895 - Method for oxidation induction test.

2. HDPE duct pipe:

S.No	Parameter	Specific Value	Specification
1.	Dimensions of the pipe		
	Size: OD/ID	40/33 mm+0.4mm	BSNL/DOT
	Wall thickness	3.5 mm+/-0.2mm	-Do-
2	Appearance	Inside & outside surface shall be smooth	
3	Colour	Green/Orange/Blue with matching lines	
4	Melt flox Index	0.2 to 1.1grms per 10 min at 190 and 5kg load	IS: 2530:1963

5	Density	940 to 958 kg/m3 at 27°C	
6.	Reversion test	3% Max	IS:4984-1995
7.	E.S.C.R (Environment Stress Crack Resistance)	Samples shall not crack and split	ASTM:D-1693
8.	IPCRT(Internal Pressure Creep Rupture Test)	Duct shall not show swelling ,leakage for 48hrs, 80°C	Is:4984-1995
9.	Crush Resistance test	Deflection not greater than 10% with load on 50kg: deflection not greater than 2% after recovery of 5 minutes	
10.	Tensile strength	20N/mm2(Min)	IS14151(part-1)
11.	Temperature Resistance & Range of product use	-20°C to +60°C	
12.	Impact Resistance	No crack and split shall be observed while dumping 10 Kg weight from 1.5 mt drop height with tup of the weight has 50,8mm dia radius edge	
13.	Coefficient of friction	Internal surface of the duct shall have less than 0.06	
14.	Standard Length of Pipe	1000 mtrs (+100)	
15.	Max outer dia of the OFC cable that can be installed by blowing technique in the HDPE pipe	16 mm	
16.	Elongation shall be	>600%	

3. Plastic (Duct) Coupler:

Size: 40 mm dia

The coupler shall be used to join two HDPE pipes. High pressure rated 10 bar air tight coupling for HDPE duct which reduces blowing resistance and allows for greater lengths of OF cable.

4. Duct Cap :

Size: 40mm dia

These shall made of rubber used for sealing the duct ends before dispatch and during transportation and laying to prevent dirt, dust, water or any other particulars from entering the duct.

5. Cable sealing plug (Simple Plug):

Size: 40 mm

Duct Inner Diametre: 33 mm (+0.2mm)

Cable outer diameter: 14 to 16 mm Simple plug with split bush assembly for sealing around cable in duct installations. Simple installation around in place cables, seals the duct from moisture, dust and dist going inside the duct in man holes.

6. RCC Route Markers

The HDPE duct shall be clearly marked at intervals of 0.2 Km with the RCC route markers of dimensions 900x300x60mm with the following information on the markers.

a) APTRANSCO OFC with Logo
b) Year of commissioning
c) Running length marking

7. Tests shall be carried on finished HDPE pipe:

- Reversion test
- E.S.C.R (Environment Stress Crack Resistance)
- IPCRT(Internal Pressure Creep Rupture Test)
- Crush Resistance test
- Tensile strength

8. Type Tests:

The HDPE pipes and accessories offered to be supplied should have been type tested either as per the requirement specified in the section 1&2 of relevant TEC specifications including latest amendments. The Bidder shall submit along with their bid the earlier carried out type test reports and the TEC certificates for the offered HDPE pipe & accessories.

9. Factory Acceptance Test:

The tests mentioned in sec -7 shall be carried out as factory acceptance for the HDPE pipe and accessories. The air pressure tests of plastic couplers shall be carried out at least for two joints of HDPE pipe.

The HDPE Duct shall be entirely suitable for laying Optical Fibre Approach Cable (OFAC) of 24F through HDPE pipe in the XLPE cable trenches. The HDPE Duct pipe shall have additive to prevent rodent attack.

The_HDPE Duct& Accessories shall also meet the following requirements:

i. Fire retardant and no acid gas evolution.

ii. Resistance to ultra-violet deterioration.

iii. Anti-moisture penetration.

10. SUPPLY: HDPE PIPE DRUMS:

The HDPE pipe shall be supplied in non-returnable strong wooden (or alternatively steel) drums provided with lagging of adequate strength, constructed to protect the cable
Against any damage and displacement during transit, storage and subsequent handling and stringing operations in the field. The bidder shall list the information concerning the following: weight, dimensions, material and standards applied.

The duct ends shall be properly sealed and secured in transport and handling.

Only one length of 1000 meters HDPE pipe shall be wound on each drum. Each drum shall be accompanied by the following information.

A. Manufacturer's name and address
B. Type of the cable
C. Gross weight of the hdpe pipe and drum
D. Net weight of the cable
E. Length of the cable
F. Drum and lot number
G. Name and address of the consignee
H. Month and year of manufacture
I. Rotation of drum

11. Installation of HDPE duct pipe:

(Excavation of trench for UG-XLPE power cable is in the scope of XLPE cable laying Contractor.)

Installation of HDPE duct and associated works shall be carried out strictly in coordination with the XLPE cable works as this HDPE duct pipe shall be laid after XLPE cable laying and the trench shall be back filled only after laying of both XLPE cable and HDPE duct pipe.

Specifications for HDPE pipe and accessories have been provided in sec 2 to 6 shall be laid at the bottom of the trench after laying of the XLPE cable and its bedding by making the surface smooth and providing 80mm stone free sand bedding. After laying the pipe additional sand shall be added to increase the height sand layer to a total of 200mm and hence positioning the HDPE duct in the middle of the sand layer.

- Line up of trench shall be such that HDPE duct will be laid in straight line, both laterally as well as vertically except at locations where it has to necessarily take a bend because of change in alignment or gradient of trench.
- Standard depth for laying HDPE pipe is at least 760 mm in normal soil and soft rock:
- Standard depth for hard rock: at least 760 mm provided the rock stars from earth crust.
- Where the berm has burrowed pits or forestation, or when cable is to be laid along culverts/bridges or cross-streams, trench may be made closer to road edge, or in some cases, over embankment or shoulder of the road.

- Ducts will be laid in a flat bottom trench, free from stones, and sharp edged debris.
- The duct would be placed in trench as straight as possible. However, at bends horizontal and vertical minimum bending radius for duct of 1300 mm would be maintained.
- Ducts will be laid preferably using specially designed dispensers.
- Ducts shall be free from twist and collapsed portions. Any such portion will be rectified before backfilling by using couplers.
- Ends of ducts will always be closed with end plugs to avoid ingress of mud, water or dust.
- Prior to aligning the ducts for jointing, each length of the HDPE duct will be thoroughly cleaned to remove all sand, dust or any other debris that may clog, disturb or damage the optical fiber cable when it is pulled or blown at a later stage.
- The ducts will be joined with couplers using duct cutter & other tools and will be tightened and secured properly.
- The duct joint will be practically airtight to ensure smooth cable blowing using cable blowing machines.
- Gl and / or RCC pipes will be used as additional protection for the HDPE ducts at rail / road crossings, built-up area/city limits, on culverts and bridges, as required.
- Chambering or concreting around RCC/ GI pipes as additional protection on bridges, culverts and also on stretches wherever depth of excavation is less than specified will be done.
- The supply & Laying of the HDPE pipe & associated accessories is the contractor's responsibility.

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11.1 BACK FILLING:

(BACK filling & Reinstatement work of excavated trench is in the scope of XLPE power cable laying contractor.)

- **Reinstatement** : Reinstatement of excavated trench will be done with proper compaction
- **THE TRENCH SHALL BE BACK FILLED ONLY AFTER LAYING BOTH XLPE POWER CABLE AND HDPE DUCT.**

- The trench shall be closed after initially filled with sieved soil or sand in rocky terrain for about 10 cm which will act as a cushion / padding and then duct is placed gently over it.
- After that another layer of 10 cm of fine sieved soil or sand is poured and then entire trench is backfilled with excavated material.
- Under normal soil conditions duct is directly laid in trench and backfill led. Adequate dry compaction will be done before crowning.

11.2 CROWNING

- When backfilling has been done up to ground level a hump of soil is made to cater for soil settlement.
- Entire excavated soil will be used for back filling.
- Crowning will be confined to width of trench only.

11.3 Duct Integrity Test

After backfilling ducts shall be tested for integrity. Two types of tests are conducted, viz., (Air Tightness Test and Kink-free Shape test).

11.3.1 AIR TIGHTNESS TEST

This shall be done by pressurizing 2 km of duct stretches at a time, by closing one end of duct and passing compressed air at 5-6kg/cm2 from the other end. When the pressure reaches about 5kg/cm2, the inlet of the duct is closed. Fall in pressure should not be more than 50% in 1(one) hour.

11.3.2 Kink-free shape test

To check that duct has not collapsed or kinked a wooden cylindrical piece (shuttle) of size 150 mm long and 0.75 x d mm in diameter, (where 'd' is inner diameter of duct), is blown into the duct with far end fitted with flexible wire grip/stocking. The wooden shuttle should pass through duct at far end without any obstruction and within approximately 10 minutes or less.

11.3.3 RCC route markers

Shall be provided and installed to mark the location and route of buried OFC/ XLPE power cables.

12. RCC JOINT CHAMBERS (MANHOLES):

SPECIFICATIONS: THE PRE CAST CYLINDRICAL RCC JOINT CHAMBERS SHALL MEET THE FOLLOWING REQUIREMENTS:

A) Size: 1200mm(bottom dia) / 950 mm (upper dia)x760mmx50mm (dia x height x thickness)
B) Upper cover :pre cast RCC 50mm thickness and diameter plate suitable for manhole covering
C) Base plate of manhole: pre cast RCC 80mm thickness and 950 mm diameter plate

13. CONSTRUCTION OF MANHOLES:

A. The RCC joint chambers (manholes) shall be provided to house joint box and optical fibre cable service loops.
B. The location for the joint boxes shall be for every 1 km of the section in the reroute.
C. The cover shall include two permanently fixed iron hooks of sufficient size for reopening purpose.
D. Minimum 3mm thickness GI strip rings shall be used inside & outside of the top edge of the manhole cylinder to avoid direct concrete contact between manhole cylinder and top cover and as well as protection of the concrete edge.
E. The bottom of the manhole shall be at the depth equal to the depth of duct (trench).

F. The manhole shall be able to with stand a live load of 20 ton vehicle and 30% impact of the moving vehicle.
G. The manholes shall include sufficient no of suitable entries for installing HDPE pipe entries and exits.
H. The manholes shall have two holes in each four perpendicular direction to HDPE pipe entries and exits.
I. Fixtures for placing cable and spliced joint box in the manhole shall be provided.
J. The joint box shall be mounted vertically on the wall of the manhole.
K. All HDPE pipe ends shall be properly sealed with duct caps or simple plugs.

14. MAINTENANCE:

- Maintenance of the HDPE pipes shall be carried out only in fault conditions or for special needs. To facilitate HDPE pipes maintenance after laying , the following shall be provided:

- The exact route descriptions, location of joint chambers shall be provided, both to

- Headquarters' office (Chief Engineer / Telecom / Vidyut soudha, Aptransco, Hyderabad) and to the terminal sub-stations.

- The Bidder shall provide/arrange any additional tools and test equipment (common and specialized) with recommended quantities necessary to install, operate and maintain all equipment to be supplied in this procurement/contract. This shall not be included/considered in the contract or evaluation of the bid.